

Automated Circuit Interpretation via Probe Prompting

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Abstract

Attribution graphs expose the circuits underlying a language model’s output, but a single graph for a short factual prompt routinely contains thousands of cross-layer transcoder features and takes on the order of two hours to interpret by hand. Existing automated interpretation methods label individual features from corpus activations, yet they do not produce circuit-level groupings calibrated to a target logit, and the labels are not validated by causal intervention. We introduce *probe prompting*, a transparent rule-based pipeline that groups the features of an attribution graph into concept-aligned supernodes from their responses on a small set of concept-targeted probe prompts, summarized as Cross-Prompt Activation Signatures (CPAS). A supernode label is *operationally useful* when intervening on the labeled supernode redirects the generated output toward the target concept (Specificity) and a structurally matched random feature set fails to do so (Matched-Control). Across five factual domains on Gemma-2-2B-it with the public CLT-HP dictionary and 42,184 entity-swap interventions, labeled supernodes satisfy both conditions in four of five domains (USA vsMax gap +5.17, Books +6.13, Products +3.23, Paintings +1.58 logits against matched random at default intervention settings), while the Sounds domain fails the matched-control condition (vsMax gap +0.14 in favor of random) and we report it as a directional negative. Secondary analyses on the same runs account for why default interventions sometimes miss: restricting the intervention to the intermediate and answer fields raises Hit-rate by 14–33 percentage points in four of five domains, and an adaptive multiplier search shows that the $M=20$ default inherited from single-feature steering overshoots in the multi-feature regime, with most rescued pairs accepting M in $[2, 7]$. Code,

datasets, and an interactive demo are released anonymously as a reusable harness for calibrating supernode labels against causal interventions.

1. Introduction

Attribution graphs operationalize the feature-to-logit pathways in a language model but do not scale to human interpretation. A single Gemma-2-2B-it graph for a short factual prompt routinely contains 600–5,000 cross-layer transcoder (CLT) features and edges between them (Ameisen et al., 2025; Lindsey et al., 2025a), and manual reading by an experienced circuit tracer has been reported to take on the order of two hours per prompt (Lindsey et al., 2025b). As mechanistic interpretability moves from individual case studies to large-scale catalogs of circuits (Hanna & Ameisen, 2026; Marks et al., 2025), automated circuit-level grouping tied to behavior is required.

Existing automated feature-interpretation pipelines (Bills et al., 2023; Paulo et al., 2024; Templeton et al., 2024) label individual features from corpus activations. The labels are per-feature, do not produce circuit-level groupings calibrated to a particular prompt’s target logit, and are not themselves validated by causal intervention on the model. What is missing is a prompt-conditioned grouping of features into circuit-level units together with a procedure that tests whether the resulting labels support the causal story they suggest.

We propose *probe prompting*, a transparent rule-based pipeline that converts an attribution graph into a compact set of concept-aligned supernodes. For each candidate feature in the graph we run a small set of concept-targeted probe prompts and summarize the feature’s responses as a Cross-Prompt Activation Signature (CPAS). Deterministic threshold rules map each signature to one of four functional roles—Semantic-Dictionary, Semantic-Concept, Relationship, Say-X—and features that share role and name are merged into a supernode. Every assignment is traceable to a specific threshold crossing, so a researcher can audit and edit the grouping.

The paper’s central evaluand is an *operationally useful* supernode label. A label for entity e is operationally useful when (i) intervening on the labeled supernode redi-

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rects the model’s output toward e (Specificity) and (ii) a structurally matched random feature set, sampled outside concept-matching supernodes with the same feature count and per-layer histogram, fails to produce the same redirection (Matched-Control). Both conditions must hold; failure on either falsifies the label. Definitions 2.6 and 2.5 fix the thresholds we use throughout: a labeled Hit-rate of at least 5 percentage points, a labeled-random Hit-rate gap of at least 5 percentage points, and a labeled-random vsMax gap of at least 1.0 logit.

Across five factual-recall domains on Gemma-2-2B-it with the public CLT-HP dictionary, labeled supernodes satisfy both conditions in four domains (USA, Books, Products, Paintings), with labeled-random vsMax gaps of +5.17, +6.13, +3.23, and +1.58 logits at the default operating point of the intervention harness. The Sounds domain fails the matched-control condition (vsMax gap +0.14 logits in favor of random) and we report it as a directional negative. The full evaluation covers 94 seed prompts and 42,184 entity-swap interventions and, to our knowledge, is the largest feature-circuit steering study reported to date; it supports both the primary test above and the diagnostic analyses of Section 6. The contemporary feature-circuit study of Hanna & Ameisen (2026) analyzes five Qwen-3 models on ≤ 100 examples per model per intervention condition.

We make no claim of full mechanistic identification of the latent variables underlying the supernodes. Our labels are behavioral abstractions with falsifiable causal consequences, at Levels 1–2 of Geiger et al. (2025); Level 3 is out of scope. All experiments use Gemma-2-2B-it with CLT-HP on English two-hop prompts, and cross-architecture, cross-dictionary, and multilingual generalization remain open.

The paper contributes:

- A transparent rule-based pipeline that groups the features of an attribution graph into concept-aligned supernodes from Cross-Prompt Activation Signatures (Section 4).
- A two-condition operational test for supernode labels, with a matched-random-control swap protocol that is grouping-method-agnostic and applicable to any future grouping, including learned ones (Sections 4 and 5).
- Cross-domain evidence that labeled supernodes pass this test in four of five factual-recall domains, plus a clearly negative fifth domain (Section 6).
- Diagnostic analyses on the same runs identifying semantic-field choice and multi-feature amplification scale as the primary drivers of intervention quality, and an open release of a 42,184-run intervention dataset, anonymous code, and an interactive demo (Appendix A).

2. Definitions

We state the definitions explicitly so that the falsifiability conditions are exposed.

Setting. Let $\mathcal{M}$ be a transformer (Gemma-2-2B-it), $\mathcal{D}$ a CLT-HP feature dictionary (Hanna & Piotrowski, 2025; Ameisen et al., 2025), and $G_p = (V_p, E_p)$ the attribution graph for prompt p with target logit y . Each node $f \in V_p$ is a (feature, position) pair with influence $\text{infl}(f) \in \mathbb{R}_{\geq 0}$; the cumulative-influence threshold τ selects a feature universe $V_p^\tau \subseteq V_p$.

Definition 2.1 (Supernode). A *supernode* $s = (\rho, \nu, F)$ is a triple of a functional role $\rho \in \{\text{SEM-DICT}, \text{SEM-CONC}, \text{REL}, \text{SAY-X}\}$, a string label ν , and a feature set $F \subseteq V_p^\tau$ all sharing role ρ and label ν . Each feature belongs to at most one supernode (Section 4).

Definition 2.2 (Concept-aligned supernode). A supernode $s = (\rho, \nu, F)$ is *aligned to concept* c if (i) ν matches c under the substring-aware matcher of Section 4, or (ii) $\rho = \text{SAY-X}$ and the discovered target token of s matches c . Each entity e in a domain induces a set of concepts $C(e)$ across its semantic fields (input, intermediate, answer); the corresponding supernodes are $S(e) = \{s : s \text{ aligned to some } c \in C(e)\}$.

Definition 2.3 (Additive entity-swap intervention). For a swap from source entity e_A to target entity e_B on prompt p , the intervention $\mathcal{I}(e_A \rightarrow e_B; M_a, M_b)$ ablates each feature $f \in \bigcup_{s \in S(e_A)} F$ by additive decoder injection with multiplier M_a (default -2) and amplifies each $f \in \bigcup_{s \in S(e_B)} F$ with M_b (default 20). Attention is *not* frozen during intervention (`FREEZE_ATTENTION=false`), so the model can route around the perturbation (Ameisen et al., 2025).

Definition 2.4 (Hit and vsMax). Let $g(p, \mathcal{I})$ be the generated continuation under intervention $\mathcal{I}$ at temperature 0.3, $n_{\text{tokens}}=10$, frequency penalty 2.0, seed 42. A swap is a **Hit** for target e_B iff the first-subword token of e_B ’s answer appears in g . The metric **vsMax** is $\max_{t \leq 10} (\ell_{e_B}(t) - \max_{e' \in \mathcal{A} \setminus \{e_B\}} \ell_{e'}(t))$, where $\ell_e(t)$ is the logit of e ’s first-subword token at trajectory position t and $\mathcal{A}$ is the domain answer set.

Definition 2.5 (Matched random control). The *matched random control* for swap pair (e_A, e_B) is an intervention $\mathcal{I}^{\text{rand}}$ identical to $\mathcal{I}(e_A \rightarrow e_B)$ in feature count and per-layer histogram, with features sampled uniformly at random from $V_p^\tau \setminus \bigcup_{e \in \mathcal{E}} \bigcup_{s \in S(e)} F$ (i.e., outside *every* concept-aligned supernode in the domain). Three deterministic replicates are drawn per pair, seeded by `sha256(run_seed : pair_id : rep : mode)`.

Definition 2.6 (Operationally useful label). The supernode label ν for entity e is *operationally useful* on prompt family p iff both conditions hold over all swap pairs (e_A, e_B) in the domain:

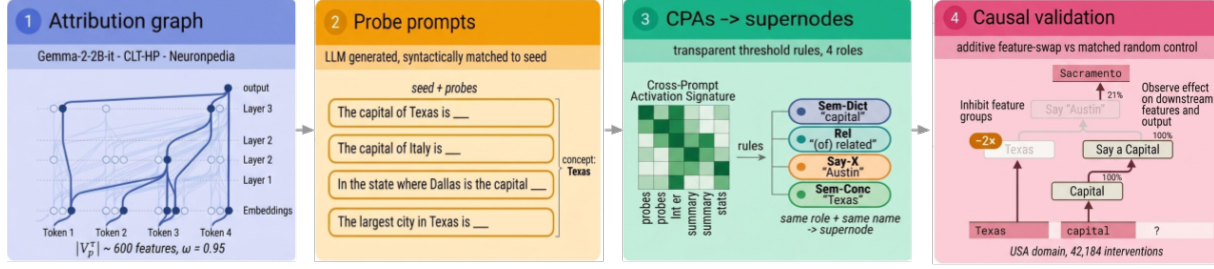


Figure 1. Pipeline overview. Four stages: (1) attribution graph generation (Neuronpedia, CLT-HP features, $\tau=0.95$); (2) probe prompts (LLM-generated, syntactically matched to seed); (3) Cross-Prompt Activation Signatures and transparent decision rules → typed supernodes; (4) causal validation via additive feature-swap interventions with structurally matched random controls (same feature count, same per-layer histogram, sampled outside concept-matching supernodes). The unit of analysis is the supernode, validated operationally by Conditions 1 and 2 of Definition 2.6.

- **Condition 1 (Specificity).** The Hit-rate of $\mathcal{I}(e_A \rightarrow e_B)$ exceeds 5 pp on a per-pair basis, and labeled vs-Max is positive on average.
- **Condition 2 (Matched-Control).** Across replicates, $\mathcal{I}^{\text{rand}}$ achieves Hit-rate at least 5 pp lower than the labeled intervention and labeled vsMax exceeds matched-control vsMax by at least 1.0 logit.

A label fails to be operationally useful if either condition fails; this is the falsification criterion used throughout the paper.

3. Related Work

Attribution graphs and replacement models. Attribution graphs operationalize feature→logit pathways in a local replacement model that linearizes the residual stream through CLT or SAE features and freezes attention (Ameisen et al., 2025; Lindsey et al., 2025a). The Neuronpedia platform exposes these graphs and per-feature cards at scale (Neuronpedia, 2025; Lindsey et al., 2025c), and circuit-tracer provides the open-source reference implementation (Safety Research contributors, 2025). Our work operates entirely *downstream* of attribution: we do not propose a new attribution method, but we do produce supernodes that the same Neuronpedia tooling can render and pin.

Sparse features and dictionary learning. Sparse Autoencoders and Cross-Layer Transcoders expose sparse, often monosemantic features that make replacement-model analyses tractable (Bricken et al., 2023; Templeton et al., 2024; Transformer Circuits Team, 2024; Marks et al., 2025). We treat the CLT-HP dictionary as fixed and study how to group its features into circuit-level units.

Feature labeling and autointerp. Automated interpretation of individual features by LLM scoring (Bills et al., 2023; Paulo et al., 2024; Templeton et al., 2024) produces

per-feature labels from corpus activations but does not yield circuit-level groupings calibrated to a target logit, and is not validated by intervention. We take the complementary route: prompt-conditioned, circuit-level grouping with explicit causal validation.

Concurrent work on causal circuits with transcoders.

Hanna & Ameisen (2026) use cross-layer transcoder feature circuits to study whether language models latently plan rhyming words, structuring their analysis around two falsifiability conditions (predict-token-before-emission, causally-affect-token). Marks et al. (2025) apply sparse-feature circuit analysis to subject–verb agreement; we contribute a domain-general harness with structurally matched random controls.

Steering with sparse features. Single-feature steering with high multipliers (Anthropic’s $M \sim 20$ paradigm) has been used to demonstrate concept-specific behavior (Templeton et al., 2024; Kharlapenko et al., 2024). We generalize to multi-feature concept-aligned swaps and find empirically that the canonical $M=20$ default is over-steering in our setting (Section 6), with the optimal M bimodal at ~ 2.4 and ~ 6.9 .

Causal interpretability and matched random controls.

Geiger et al. (2025) formalize three levels of interpretive claim and emphasize that *distinguishing* levels requires intervention. Heap et al. (2025) show that SAE auto-interpretability scores (Bills et al., 2023; Paulo et al., 2024) and standard SAE reconstruction metrics can be similar for randomly-initialized and trained transformers, highlighting the need for metrics anchored to causal effect. Our matched-random-control protocol (Definition 2.5) is designed to address this critique: any structural property a random feature set shares with the labeled one (count, layer histogram, eligibility) is held constant; only the concept alignment changes.

4. Method

4.1. Setting and attribution-graph generation

We use Gemma-2-2B-it with the public CLT-HP dictionary ($\sim 2.5\text{M}$ features over 26 layers; Hanna & Piotrowski, 2025; Ameisen et al., 2025). Attribution graphs are generated via the Neuronpedia API with a relatively permissive node-inclusion threshold, a high edge-influence threshold, and a cap on the number of feature nodes per graph; exact API parameter names and values are in Appendix B. From each graph we select a feature universe V_p^τ by cumulative-influence threshold $\tau=0.95$ for batch experiments. The replacement model freezes attention during graph computation; downstream interventions (Definition 2.3) run with attention free, making positive results harder than under frozen attention (Ameisen et al., 2025).

4.2. Probe prompt generation

Given a seed prompt p (e.g., “the capital of the state containing Dallas is”), we use an LLM in two passes to generate a small set of concept-targeted probes. The first pass extracts the K most salient concepts from p (and, when available, from the model’s output), each tagged with a category (entity, process, relationship, attribute) and a short description. The second pass turns the extracted concepts into probe prompts that share the seed’s surface structure (same prepositions, same answer position, comparable length) while varying the content entity. A typical seed yields 20–30 probes. The full system prompt and the concept-to-probe template are reproduced in Appendix B.

4.3. Cross-Prompt Activation Signatures (CPAS)

For each (feature, probe) pair we record a small set of per-probe measures: which token receives the feature’s peak activation and where in the sequence it occurs, how sparsely the feature fires across the probe set, and a robust z -score and cosine similarity of the feature’s activations relative to the seed prompt. Aggregating across probes yields a compact per-feature signature (the CPAS): a handful of summary numbers capturing how consistent the peak token is across probes, how many distinct tokens the feature peaks on, the balance between peaks on functional and semantic tokens, and the feature’s confidence in its dominant role. Formal definitions and raw-measure specifications are in Appendix B.

4.4. Functional vs. semantic tokens; target-token mapping

Tokens are labeled *functional* (copulas, articles, prepositions, conjunctions, relative pronouns, auxiliaries) or *semantic* (content-bearing). When a feature peaks on a functional token, a ± 7 -token directional search identifies the near-

est semantic target (e.g., forward for *is/are/articles/most* prepositions; backward for *of/possessive 's*; bidirectional for conjunctions/punctuation). The full vocabulary and edge cases are in Appendix C.

4.5. Four functional roles

Each feature in V_p^τ is assigned one of four functional roles. The four-role vocabulary is an empirical extension of the supernode types that appear in circuit-tracing case studies (Lindsey et al., 2025a; Ameisen et al., 2025); we found that these four roles covered the vast majority of features we encountered during pilot analyses, and we formalize them here as the pipeline’s output types.

- **Semantic-Dictionary (SEM-DICT).** The feature fires consistently on the *same* semantic token across probes—a dictionary-like detector for a specific concept (e.g., a feature that reliably peaks on “Texas” whenever Texas is mentioned).
- **Semantic-Concept (SEM-CONC).** The feature peaks on semantic tokens across a small family of related tokens rather than on a single one—a concept-level rather than token-level detector, typical of middle layers.
- **Relationship (REL).** The feature does not concentrate its activity on any particular token; it fires diffusely across the probe, with a comparatively dense activation pattern. These features appear to encode a relation or context rather than a named entity.
- **Say-X (SAY-X).** The feature peaks on functional tokens (e.g., *is, the*) in predictable positions relative to a semantic target, and sits in the later half of the network. After the ± 7 -token directional search (Section 4.4), the feature is named by the *target* semantic token that follows or precedes the functional peak.

A strict-priority decision tree on the CPAS assigns each feature to one of these roles; a fifth bucket (REVIEW) captures features that do not match any rule confidently, and is excluded from downstream analysis. The exact thresholds and priority ordering are in Appendix B. We chose rules rather than a learned classifier because the grouping itself is the object of interpretability research—every assignment maps back to a specific threshold crossing that can be audited and edited by a researcher.

4.6. Supernode formation and naming

Same-role same-name features form a supernode (Definition 2.1). Cross-prompt stability is enforced ($\geq 60\%$ of probes); features failing this are marked *ungrouped* (5–10% of candidates) and excluded. Naming is role-specific.

Semantic features (both SEM-DICT and SEM-CONC) are named by the semantic token at which they activate most

strongly: we filter the feature’s per-probe records to those where the peak is on a semantic token, sort by activation strength, and select the highest-activation token that is not in a configurable blacklist. A feature that consistently peaks on “Texas” receives the name *Texas*; if no semantic peaks are found, the system falls back to the corresponding token in the seed prompt.

Say-X features are named by the semantic token they appear to promote. Since these features peak on functional tokens, the naming uses the semantic target discovered during the directional search of Section 4.4; the final format is *Say (token)*, e.g., *Say (Austin)* for a feature peaking on *is* immediately before “Austin”.

Relationship features are named through a two-stage process. We first build an extended semantic vocabulary combining tokens from the seed prompt with concept names already discovered for semantic features; we then aggregate activation across all probes for each semantic token and pick the highest-activation entry not in the blacklist. The format is *(token) related*, producing names like *(containing) related* for features that fire diffusely on spatial-relationship phrases.

4.7. Pipeline output

The pipeline output is a property of the method. Concept-aligned subgraphs compress 600+ raw features into 30–50 named supernodes per circuit while preserving Neuronpedia *Completeness* (mean 0.83 over the five illustrative circuits) and accepting a *Replacement* drop to 0.54. The trade-off is intentional: low-influence nodes are treated as error to maximize legibility. The unit of analysis in Section 6 is the supernode.

4.8. Causal-validation harness

Intervention mechanism. Additive delta injection via CLT decoder vectors (Definition 2.3). For each intervened feature, $v_{\text{new}} = M \cdot v_{\text{orig}}$ scales the decoder vectors and adds the result to the residual stream at downstream layers. Defaults: $M_{\text{ablate}} = -2$, $M_{\text{amplify}} = 20$. Attention is not frozen during intervention; positive results are stronger evidence than under frozen-attention patching where some direct effects are forced by construction (Ameisen et al., 2025).

Matched random control (Definition 2.5). For each swap pair we construct a random feature set with the same feature count and per-layer histogram as the labeled intervention, drawing features from V_p^T outside every concept-aligned supernode in the domain so that the only property that varies is concept alignment. Three deterministic replicates per pair yield Hit-rate and vsMax distributions over random feature sets. The protocol is grouping-method-

agnostic and can be re-run on any competing grouping, including future learned ones.

Field-additivity decomposition. Each domain has 3 semantic fields (input, intermediate, answer; see Section 5). For each pair, 7 variants are run: 3 single-field, 3 two-field, 1 all-three. Each selected field drives both ablation and amplification; this isolates which fields carry the causal signal.

Adaptive M -search. When a swap misses at the default $M_{\text{amplify}} = 20$, we search for a better amplifier in two phases. Phase 1 probes a coarse geometric grid on $[0.1, 20]$ (6 points) and stops at the first hit. If Phase 1 finds no hit, Phase 2 uses the fact that the steering effect typically has a sharp onset: we compute the KL divergence of the steered output against the unsteered baseline at each probed M , locate the interval in which KL rises most steeply, and binary-refine inside that interval for 6 further steps, accepting a hit at any refinement point. The rationale is that hits tend to cluster near the onset of a meaningful model response. Applied to labeled and field-additivity runs (1,210 + 867 eligible pairs); full pseudocode in Appendix F.

4.9. Metrics

We use three primary metrics (per Definition 2.4): **Hit%**, the fraction of pairs whose generated continuation contains the target’s first-subword token; **vsMax**, the maximum over the generated trajectory of the target’s logit minus the best other answer in the domain (positive means the target is ahead); and **Target Recovery**, the fraction of pairs in which the target’s logit at some trajectory position exceeds its own unsteered baseline, indicating the intervention actually moved the target-direction and did not just suppress the source. A small set of secondary diagnostics (suppression of the source, target–source rank crossover, rank of the target within the domain answer set, a control-token stability measure, and KL at position 0) is described in Appendix B.

5. Experimental Design

5.1. Domains and seed prompts

We evaluate on five domains spanning two-hop factual recall in different relational structures and answer geometries. Each domain has a fixed seed-prompt template with three semantic fields (input, intermediate, answer): USA (“capital of state containing X ”; 50 entities, 2,450 non-identity swap pairs, answer-set size $K=50$), Books (“character X from book by”; 16/240/25), Products (“product X from company by”; 12/132/25), Paintings (“painting X by painter”; 10/90/20), and Sounds (“sound X from animal of color”; 6/30/6). The first four are the *strong* domains used for cross-domain inference; Sounds has 12/30 pairs sharing an answer and is reported separately as a directional negative

(Section 6.3, Section 7).

5.2. Experimental matrix

Each swap pair is run under: (a) labeled (3-field) baseline at $M=20$; (b) matched random control (3 replicates per pair); (c) field-additivity (7 variants per pair); (d) adaptive M -search on labeled and on the best field-add variant per missed pair. The total is **42,184 swap runs** across all conditions, domains, and M values.

5.3. Pre-registered evaluation choices

Before running the sweep we fixed the evaluation protocol. The contrast set used by vsMax is the set of all other dataset answers in the same domain, and multi-token answers are resolved to their first subword so that Hit is a deterministic substring check. Intervention and measurement are explicitly decoupled: the intervention operates on supernodes tied to entity fields, while the Hit/vsMax readout is always taken on the answer field. Generation uses temperature 0.3, a maximum of 10 new tokens, a frequency penalty of 2.0, and a fixed seed; matched-control replicates are drawn with sha256-based seeding from the run seed and pair identifier. At the pair level we count a random-hit if any of the three matched-control replicates is a hit, so the random baseline is deliberately optimistic. Ranges, thresholds, and seeds were frozen prior to the 42,184-run sweep. Full details are in Appendix B.

6. Results

The evidence is organized in three blocks. Section 6.1 tests Definition 2.6’s two conditions using a canonical intervention protocol. Section 6.2 uses the same 42,184 runs to analyze the two primary levers that move a pair from miss to hit (semantic-field composition and amplification scale) and two structural correlates of domain-level steerability (cross-prompt feature overlap and graph-scaffold compatibility). Section 6.3 records the one domain in which the primary test fails and one within-domain null.

6.1. Primary evaluation

We evaluate operational usefulness against Definition 2.6 using the intervention harness of Section 4.8. For each source–target pair (e_A, e_B) , the harness constructs an additive entity-swap on the concept-aligned supernodes of both entities and reports two operating points. The *default* operating point fixes all three semantic fields (input, intermediate, answer) with ablation multiplier $M_{\text{ablate}} = -2$ and amplification multiplier $M_{\text{amplify}} = 20$; the *harness* operating point additionally selects the best field subset for the pair and accepts any amplification value identified by the adaptive M -search. The matched random control of Definition 2.5

is evaluated at the same operating point as its labeled counterpart. Because labeled and random interventions share feature count and per-layer histogram by construction, any systematic gap between them reflects concept alignment.

At the default operating point, labeled supernodes produce a positive vsMax against matched random in every strong domain: +2.86 against -2.31 in USA, +5.98 against -0.15 in Books, +3.46 against +0.23 in Products, and +1.55 against -0.03 in Paintings (Appendix D), so the labeled–random vsMax gap ranges from +1.58 to +6.13 logits across the four strong domains. At the same default, labeled Hit-rate also exceeds matched random: USA 24.7 vs 0.1%, Products 15.2 vs 0.3%, Books 3.8 vs 0.3%, Paintings 4.4 vs 0.0%. In Books and Paintings the default labeled Hit-rates fall below the absolute 5 pp threshold of Condition 1. At the harness operating point, labeled Hit-rate rises to 67.3, 63.8, 27.6, and 21.8% in USA, Books, Products, and Paintings, while matched random stays below 2% in every domain (Table 1), and the absolute threshold of Condition 1 is met in all four. Sounds inverts the vsMax ordering: labeled vsMax is +3.28 against matched random’s +3.14, a gap of only +0.14 logits.

Raw source suppression does not discriminate labeled from random. Matched random controls achieve equal or higher suppression of the source answer than labeled interventions in three of five domains—Books 75 vs 69%, Products 87 vs 65%, Paintings 74 vs 34% (Appendix D)—yet redirect the output nowhere in particular. A per-swap disruption-regime taxonomy at position 0 confirms the distinction: matched random concentrates in a generic-disruption regime (42–45% of USA and Paintings pairs), and the labeled harness concentrates in a clean-redirection regime (Appendix K). Within the mixed regime in which both target and source logits drop but the target overtakes, the labeled intervention drives the target above its own unsteered baseline on 92.2% of USA pairs against 29.3% for matched random; sustained target dominance is 0.673 against 0.319, and within-regime vsMax is +2.33 against -0.10 (Appendix K). The discriminating signal is therefore targeted redirection, and targeted redirection is a property of the labeled feature sets.

Bootstrap 95% confidence intervals on Hit-rate gaps and permutation tests on overlap statistics are reported in Appendices D and I. With 2,450 USA pairs and 240 Books pairs, the observed labeled–random Hit-rate gaps of 67.1 and 63.8 percentage points lie far outside any plausible null under the matched-random distribution.

The four strong domains satisfy both conditions of Definition 2.6. The labeled–random vsMax gap at the default operating point exceeds 1.0 logit in every one (+5.17, +6.13, +3.23, +1.58 logits), satisfying the vsMax component of Condition 2 and the labeled–vsMax-positive component of Condition 1. At the harness operating point, the absolute

Table 1. Headline (C1): labeled supernodes vs. matched random controls. For each domain we report (a) the per-pair best field-add variant with adaptive M -search Hit% (a comprehensive operational measure), and (b) the labeled vs. matched-random vsMax discriminator at the full-labeled all-fields $M=20$ baseline (the cleanest specificity signal; full per-condition tables in Appendix D). Hit% counts pairs whose first generated subword matches the target answer; vsMax is the best logit margin against the strongest competing dataset answer. C1 (Definition 2.6) is supported when both Hit-rate gap ≥ 5 pp and vsMax gap ≥ 1.0 logit. Sounds fails Condition 2 (vsMax gap $+0.14$) and is reported as a directional negative (Section 6.3).

Domain	Hit% (best-FA, M -opt)			vsMax (full lab. vs. rand., $M=20$)			Verdict
	Lab.	Rand.	Δ	Lab.	Rand.	Δ	
USA	67.3	0.2	+67.1	+2.86	-2.31	+5.17	C1 supported
Books	63.8	0.0	+63.8	+5.98	-0.15	+6.13	C1 supported
Products	27.6	1.1	+26.5	+3.46	+0.23	+3.23	C1 supported
Paintings	21.8	0.0	+21.8	+1.55	-0.03	+1.58	C1 supported
Sounds	20.0	3.3	+16.7	+3.28	+3.14	+0.14	<i>Negative</i> (Section 6.3)

labeled Hit-rate exceeds 5 pp in every one (67.3, 63.8, 27.6, 21.8%) and the labeled-random Hit-rate gap exceeds 5 pp in every one (+67.1, +63.8, +26.5, +21.8 pp), satisfying the Hit-rate components of both conditions. Sounds fails the vsMax component of Condition 2 and is reported in Section 6.3.

6.2. Diagnostic analyses

The analyses below use the same 42,184 runs to characterize variation in intervention quality. Each is a follow-up to the primary test of Section 6.1; the four-domain verdict above does not depend on any of these diagnostics individually.

6.2.1. SEMANTIC-FIELD COMPOSITION

Restricting the intervention to a two-field subset substantially improves redirection over the default all-three-field configuration. In USA the best subset (*state + capital*) raises Hit% from 24.7 to 38.8; in Books (*book + author*), from 3.8 to 37.1; in Products (*company + founder*), from 15.2 to 24.2; in Sounds (*sound + animal*), from 0.0 to 20.0. Paintings has a weaker baseline and a correspondingly weaker gain (best subset *first_name*, 3.3 to 6.7). Aggregated across the five domains, the intermediate field alone drives Hit% 14.6%, the answer field 8.8%, and the input field only 6.1% (Appendix E). Input-field features appear to encode the model’s representation of the surface form, while the circuits that select the emitted answer concentrate in the intermediate and answer fields; including input-field features dilutes the redirection signal with activity that is not aligned with the target answer.

6.2.2. AMPLIFICATION SCALE

The $M=20$ amplification default inherited from single-feature steering overshoots in the multi-feature regime. Adaptive M -search rescues 452 pairs across all conditions, and the distribution of winning values is bimodal, with peaks near $M \sim 2.4$ (47% of labeled hits) and $M \sim 6.9$ (40%), both below the default (Appendix F). Two controls rule out the

alternative that the default is simply too small. Testing $M \in \{50, 100, 200\}$ on 80 near-miss pairs across three domains produced 0 additional hits; KL divergence between the steered and unsteered distributions saturates above $M \sim 50$ while the generated text continues to degrade. Testing the top- k features by graph influence ($k \in \{1, 3, 5, 10\}$) at $M \in [20, 200]$ on Products produced 0 hits out of 96, and only the full feature set yields hits. The steering signal is distributed across the feature set, and multi-feature amplification at high M linearly amplifies every direction in the labeled set simultaneously, producing distributed noise that dilutes the target signal. The single-feature SAE setting of Templeton et al. (2024), in which clamping at roughly $\pm 10 \times$ maximum observed activation saturates one direction, behaves differently for a structural reason: a single direction cannot linearly superpose with itself.

6.2.3. CROSS-PROMPT FEATURE OVERLAP

A scalable cross-prompt analysis on 1,607 intra-domain entity pairs spanning 103 entities characterizes how each entity’s feature set distributes across the 26 layers of Gemma-2-2B-it (Appendix I). Activation stability—the probability that a feature active on the seed prompt is also active on a concept-matched probe for the same entity—exceeds 90% in all domains, so the CPAS picture of each entity is reproducible at scale. Jaccard overlap between entity feature sets within a domain ranges from 0.29 to 0.62, all significantly above chance under permutation ($p < 0.001$ at several pool sizes), and the ratio of early-layer to late-layer Jaccard is between 1.4 and $3.0 \times$ in four of five domains. The pattern is consistent with a shared backbone of early-layer features reused across entities and entity-specific specialization concentrated in later layers.

6.2.4. SCAFFOLD COMPATIBILITY

Scaffold compatibility tracks cross-domain steerability. We define per-domain *scaffold compatibility* as the fraction of total graph influence carried by features that receive iden-

tical supernode assignment across two entity graphs. With one scaffold number per domain, the metric orders the four strong domains by Hit% at the default operating point in the same sequence (Spearman $\rho=1.0$, $N=4$; Appendix J): USA (scaffold 0.53, Hit% 24.7), Products (0.42 $\rightarrow$ 15.2), Paintings (0.36 $\rightarrow$ 4.4), and Books (0.25 $\rightarrow$ 3.8). Restricting to later layers ($\geq L_{16}$) makes the gradient steeper. The cross-domain ordering does not carry down to a pair-level predictor within a single domain: in USA, splitting the 2,450 pairs into quartiles of pair-level scaffold compatibility gives nearly flat Hit-rates (27.8, 24.7, 24.3, 24.3%) with absolute Pearson correlations below 0.10 for every variant tested, plausibly because the within-USA scaffold range is narrow ([0.38, 0.73]). In smaller domains the within-domain signal is weak but non-zero (e.g., residualized $r = +0.29$ in Products with $N=132$). We therefore limit the scaffold claim to the between-domain level and report the within-USA null in Section 6.3.

Collectively, these diagnostics account for the gap between the default and harness operating points. Semantic-field composition and amplification scale are the two levers that move a pair from miss to hit under the same labeled feature set, and the cross-prompt overlap and scaffold analyses locate a structural correlate of domain-level steerability.

6.3. Negative results

We record two negatives alongside the primary evaluation.

Sounds fails Condition 2 of Definition 2.6. The domain has six entities, and 12 of the 30 non-identity pairs share the target answer “brown”. Matched random controls that suppress the source color produce the target color mechanically on the shared-answer subset, inflating random Hit% to 12.2% at the harness operating point; the matched-control vsMax gap is +0.14 logits in favor of random. We attribute the failure to small N , shared answers, and the thin logit margins induced by the domain’s $K=6$ answer set (Appendix M) and exclude Sounds from cross-domain inference for the primary claim.

The scaffold metric does not predict pair-level Hit-rate within USA, our most statistically powered domain. Splitting the 2,450 USA pairs into four quartiles of pair-level scaffold compatibility gives Hit-rates of 27.8/24.7/24.3/24.3%, essentially flat, with $|r| < 0.10$ for all variants tested. The likely reason is that within-domain scaffold values in USA lie in the narrow range [0.38, 0.73], too narrow to separate pairs. The scaffold claim is therefore between-domain only, and we do not use scaffold compatibility as a within-domain predictor.

7. Limitations and Threats to Validity

1. No grouping-method baseline. We do not compare against geometric clustering (cosine, layer-adjacency, k -means, HDBSCAN). A grouping derived from a behavior-grounded signature outperforms a purely geometric grouping on behavior-grounded metrics nearly by construction, so such a comparison would be structurally biased in our favor; and there is no canonical feature-clustering baseline in the SAE/CLT literature to anchor the comparison. We validate instead via intervention against matched random controls (Definition 2.5), which is method-agnostic and applicable to any future grouping, including learned ones, and we encourage comparisons on our harness.

2. Attention-circuit blindspot. Attribution graphs freeze attention; our swaps run with attention free. The asymmetry plausibly explains part of the 33–79% miss rate in the strong domains (Ameisen et al., 2025). Attention-aware variants are listed as future work (Section 8).

3. Sounds dataset structural issues. Six entities, 12/30 pairs share answers, $K=6$. Sounds fails Condition 2 of Definition 2.6 (Section 6.3) and is excluded from cross-domain inference for the main claim.

4. Substring matching and multi-token heuristics. The concept-to-supernode matcher uses bidirectional substring matching, which allows reverse containment (e.g., the short supernode name `is` matching the longer concept `mississippi`, or `port` matching `gulfport`). We audited this behavior and found that entities with the highest reverse-substring rates still perform at or above their dataset mean on vsMax (for example, the USA entity with the highest such rate, Mississippi/Gulfport, has vsMax 3.20 vs a USA mean of 2.86), and that within-entity field-additivity does not show a disproportionate drop when reverse-substring matches are added. The labels used in the main-paper experiments were manually inspected against the generated supernode names for every seed; the few cases we found to be unambiguous mismatches were corrected before the sweep. Multi-word answers resolve to their first subword, which can miss or produce spurious matches (e.g., “St. Paul” vs stored “Saint Paul”). Tightening the matcher is a first-pass improvement for future work.

5. Threshold sensitivity is unmeasured. Decision-rule thresholds were iteratively refined on pilot circuits; we did not run a formal ± 10 –20% sensitivity sweep. Thresholds are treated as configurable defaults in the released code.

6. Cross-prompt robustness is observational. The cross-prompt feature-overlap pattern (§6.2.3) comes from measuring feature-set overlap across 1,607 entity pairs; we did

not then *intervene* on shared-early-layer features to verify that ablating them damages many entities uniformly, nor on specialized late-layer features to verify that ablating them damages only one entity. The Jaccard pattern is consistent with a shared early-layer backbone and late-layer specialization, but that reading remains correlational until tested by intervention.

7. Within-domain pair-level scaffold prediction is null in USA. Scaffold compatibility orders our four strong domains by Hit% exactly at the between-domain level ($N=4$) but does not predict Hit-rate at the pair level within USA, our most statistically powered domain (Section 6.3). The within-domain scaffold range $[0.38, 0.73]$ for USA is probably too narrow to discriminate pairs; we therefore limit the scaffold claim to the between-domain level.

8. Three levels of interpretive claim; single model and language. In the three-level framing of Geiger et al. (2025)—Level 1 operationally useful labels, Level 2 downstream causal effects validated against controls, Level 3 full mechanistic identification of latent variables—we claim Levels 1 and 2 only; Level 3 is disclaimed. All experiments use Gemma-2-2B-it with CLT-HP and English two-hop prompts (Section 5.1); cross-architecture, cross-dictionary, and multilingual generalization are open.

8. Discussion and Conclusion

Methodological lessons. Three findings in this paper may inform intervention practice in related settings. First, matched random controls need to be *structurally* matched—same feature count, same per-layer histogram, and sampled outside every concept-aligned supernode—because otherwise the labeled–random gap is easy to misstate: in three of our five domains, matched random feature sets produce *more* source-suppression than the labeled intervention does, even though they redirect the model nowhere in particular. Second, the $M=20$ amplification default that is standard for single-feature steering does not carry over to the multi-feature CLT setting studied here: our winning values of M are bimodal at roughly 2.4 and 6.9, and a coarse sweep over M rescues a substantial fraction of misses. Third, semantic fields are not interchangeable for intervention purposes: restricting to intermediate- and answer-field features gains 14–33 pp of Hit-rate over using all three fields in four of five domains, consistent with input-field features encoding the model’s representation of the surface form and the answer-selection circuits living in the intermediate and answer fields.

What the labels mean. Supernode labels are behavioral abstractions: they pass an operational test against matched controls without amounting to mechanistic identifications

of latent variables in the full Level-3 sense of Geiger et al. (2025).

Polysemanticity and cross-prompt activation. CLT and SAE features are not guaranteed to be monosemantic in practice. Templeton et al. (2024) document *feature splitting*—a coarse feature in a smaller dictionary splits into several finer features as dictionary capacity grows—indicating that “one feature” is often a bundle of related meanings. Lindsey et al. (2025a) note this explicitly when constructing Michael Jordan supernodes: some member features are polysemantic in a dictionary-wide sense, and are kept in the supernode only on account of the facet relevant to the traced prompt. Balcells et al. (2024) document related pass-through and specialization phenomena layer-by-layer. Probe prompting formalizes this ad-hoc practice. By comparing a feature’s activation across concept-targeted probes, CPAS assigns each feature a *prompt-contextual* role: the feature is monosemantic enough, in the context of the prompt family under study, to serve as the unit of a supernode, even when it is polysemantic across the dictionary as a whole. The same logic suggests that the method is not strictly tied to a trained sparse dictionary and can in principle be applied to raw MLP neurons, which are known to be substantially more polysemantic (Anonymous, 2026): when neurons are inspected on a small set of carefully chosen probes instead of on corpus-wide activations, a usable functional signature can emerge even without the sparsity prior.

Future work. Several directions seem natural. An obvious first extension is to make the intervention attention-aware, either by incorporating edge-attribution information or by also swapping attention-head outputs along with CLT feature activations; replicating the main findings on Gemma-2-9B, Llama-3, and alternative feature dictionaries would test how much of the current picture is specific to Gemma-2-2B-it and CLT-HP. A second direction is multilingual extension: our current prompt templates are English-only, and we have observed (Appendix C) that the functional-token vocabulary and the directional search it drives do not transfer cleanly to, for example, French. A third direction is to strengthen the within-domain scaffold predictor; the cross-domain signal is clean, but the within-domain null in USA remains an open research question.

Two further directions deserve specific mention. The first concerns the *probe set* itself. Our current probes are positive variations on the seed prompt (“the ⟨entity⟩ is”), and we treat the set as fixed across features. There is no reason to do so. Contrastive or near-miss probes—probes that differ minimally from the seed along a concept dimension the model might confuse—should be considerably more discriminating. At longer context lengths, standard for frontier models, one can imagine agentic variants of the harness

that treat probe design as a detective game: an agent with a bounded action budget chooses which substring or position to intervene on next, based on what the previous probe revealed. The second direction concerns the feature-role taxonomy itself. The four-role vocabulary (Semantic-Dictionary, Semantic-Concept, Relationship, Say-X) is an empirical extension of the roles that appear in circuit-tracing case studies (Lindsey et al., 2025a), and the fact that it generalizes across five unrelated domains suggests there is a useful “model-biology” object hiding behind it; sharpening this taxonomy—adding roles, subdividing existing ones, or reducing them—is a research question in its own right. Finally, the grouping machinery itself is open: learned CPAS classifiers trained on the rule-based decisions could replace the hand-authored decision tree with a smoother map from signatures to roles, and CPAS with $\delta \times \text{grad}$ candidate selection can be used on dense MLP neurons to profile circuits outside the SAE/CLT setting (Anonymous, 2026).

Conclusion. Probe prompting shows that circuit-level interpretability of attribution graphs can be partially automated with a transparent, rule-based pipeline, and that the resulting labels can be held to a two-condition operational test against matched random controls. The test is sharp enough to falsify labels in one of our five domains, and the harness produces substantive findings about field interference, over-amplification, and scaffold compatibility that inform practice in the steering literature.

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A. Reproducibility Statement

To support the workshop’s emphasis on reproducibility and code/data access, we release the following anonymously for the review period.

- **Code.** Anonymous repository at <https://anonymous.4open.science/r/attribution-graph-probing-anon> (commit hash and license in README.md).
- **Datasets.** The full 42,184-run swap dataset (graphs, prompts, activations, supernode groupings, swap dumps for all five domains) at the same anonymized URL, with a per-domain manifest in Appendix N.
- **Interactive demo.** An anonymized HuggingFace Space at <https://anonymous.4open.science/r/attribution-graph-probing-demo-anon> that accepts an attribution graph and returns concept-aligned supernodes with the matched-control diagnostics.
- **Per-domain CLI examples.** Appendix N gives end-to-end commands for the USA, Books, Products, Paintings, Sounds pipelines, including the exact τ , M , and seed used for each table and figure.
- **Determinism.** The pipeline is fully deterministic (sha256-seeded random controls; fixed temperature/seed/penalty for generation; no random initialization); per-feature checkpoints permit resumption from partial runs.
- **Hardware.** A single L4 GPU is sufficient to reproduce any single domain in 14–24 minutes (graph generation + activation measurement + grouping + subgraph), excluding the swap sweep; the full swap sweep is documented per-domain in Appendix N.

We commit to de-anonymizing the URLs at camera ready and to maintaining the artifacts indefinitely.

Appendix

The appendix is organized to track the main paper: §B–§C give a detailed reference for the pipeline and its inputs; §D–§E back up the results of §6.1 with per-domain numbers; §F–§H back up the M -search and steering-strength discussion of §6.2.2; §I–§J back up the cross-prompt and scaffold analyses of §6.2.3 and §6.2.4; §K gives the logit-shift regime taxonomy used throughout; §L contains worked case studies; §M analyzes the two negative-result domains (Sounds and Paintings); §N is the reproducibility manifest; §O is the glossary. Every appendix section is self-contained and cross-referenced from the main text.

B. Pipeline Reference

B.1. End-to-end pipeline

The pipeline takes an attribution graph for a seed prompt and returns a collection of concept-aligned supernodes together with the data needed to run matched-control swap interventions. The pseudocode below summarizes the steps; the paragraphs that follow describe each step in plain language.

```

Input:  attribution graph G_p, seed prompt p, target logit y
Output: concept-aligned supernodes {S_i}, swap harness I

1. V_tau = select_features_by_cumulative_influence(G_p, tau=0.95)
2. C     = llm_generate_concepts(p, K)
3. Probes = synthesize_probes(C, template_constraints)
4. for each feature f in V_tau, each probe q in Probes:
50    A[f, q] = measure_activation(f, q)
51. CPAS[f] = aggregate_cross_prompt(A[f, *])
52. for each feature f in V_tau:
53    role[f] = classify(CPAS[f], thresholds)
54    name[f] = assign_name(f, A, blacklist, target_token_rules)
55. Supernodes = group_by((role, name)), stability >= 0.6
56. for each swap pair (e_A, e_B):
57    I_lab = build_swap(S(e_A), S(e_B), M_ablate=-2, M_amplify=20)
58    I_rand = matched_random_control(I_lab)
59    run(I_lab); run(I_rand)
9. adaptive_M_search(missed_pairs)

```

Step 1 (feature selection). Attribution graphs from Neuronpedia are generated with a relatively permissive node-inclusion threshold (`nodeThreshold=0.8`), a high edge-influence threshold (`edgeThreshold=0.85`), and a cap of `maxFeatureNodes=5000`. The graph typically contains 600–5,000 nodes; we then retain only the nodes whose cumulative influence sums to at least $\tau=0.95$, which reduces each circuit to 200–700 features for batch experiments.

Step 2–3 (probe generation). The seed prompt is handed to a separate LLM together with a small specification that (a) lists the *concepts* we expect to find in the graph (entities, relations, answer tokens) and (b) constrains the probe templates to be syntactically close to the seed (same prepositions, same answer position, same length). The LLM returns a handful of concept-targeted probe prompts. Typical probe counts are 20–30 per seed.

Step 4–5 (activation measurement and CPAS). Each feature in V_τ is re-run on each probe and its activation measured with the Neuronpedia API. The per-(feature, probe) record stores the peak token, the peak position in the probe, the activation density, a sparsity ratio, a robust z -score, and a cosine similarity to the seed activation. These raw measures are then aggregated across probes into a compact per-feature signature (the CPAS).

Step 6 (classification and naming). A strict-priority decision tree (§4.5; thresholds in Table 3) assigns each feature one of four functional roles (Semantic-Dictionary, Semantic-Concept, Relationship, Say-X) or a fifth REVIEW bucket that is excluded from downstream analysis. Naming is role-specific (§4.6).

Step 7 (supernode formation). Same-role same-name features are merged into a supernode; features whose role or name is inconsistent across probes ($< 60\%$ probe-level stability) are marked *ungrouped* and excluded.

Step 8–9 (causal-validation harness). For every entity pair in the domain we build an additive entity-swap intervention (Definition 2.3), run it at $M_{\text{ablate}} = -2 / M_{\text{amplify}} = 20$ with attention *not* frozen, and compare it to three deterministic matched-random-control replicates (Definition 2.5). Pairs that miss at the defaults enter adaptive M -search (§F).

B.2. CPAS measures

The CPAS is the seven-number summary that the decision tree operates on. Table 2 gives the formal definitions and the intuition for each field.

Field	Definition and intuition
<code>peak_consistency_main</code>	Fraction of probes whose peak token matches the feature’s modal peak. High (≥ 0.8) means the feature is a dictionary-like detector for one particular token.
<code>n_distinct_peaks</code>	Number of distinct peak tokens across <i>active</i> probes. = 1 for a Semantic-Dictionary feature; larger for Semantic-Concept or Say-X.
<code>share_F / conf_F</code>	Share of active probes whose peak falls on a functional token / classifier confidence on that share. A feature with <code>share_F</code> ≈ 1 and high <code>conf_F</code> is a Say-X candidate.
<code>func_vs_sem_pct</code>	$100 \cdot (\max_{\text{func}} - \max_{\text{sem}}) / \max_{\text{overall}}$. Positive when the peak is on a functional token; used to break ties between Say-X and semantic roles.
<code>sparsity_median</code>	Median of $(\text{peak} - \text{mean}) / \text{peak}$ across active probes. Low values indicate a dense, diffuse activation pattern characteristic of Relationship features.
<code>conf_S</code>	$1 - \text{share}_F$, i.e. confidence that the peak is on a semantic token.

Table 2. CPAS fields used by the classification rules in §4.5.

B.3. Decision-rule thresholds

The decision tree is a strict-priority cascade: features are evaluated against the rules in the order below, and the first matching rule wins. Thresholds were iteratively refined on a small set of pilot circuits and then frozen before the 42,184-run sweep.

The layer-based asymmetry (Say-X requires layer ≥ 7 , Sem-Conc accepts layer ≤ 3) encodes a rough prior observed in Gemma-2-2B-it: early layers are overwhelmingly semantic-content features; the “Say X” behavior of functional-token features promoting a specific answer is observed only in the later half of the network.

C. Functional Vocabulary and Target-Token Mapping

The pipeline distinguishes *functional* tokens (copulas, articles, prepositions, conjunctions, relative pronouns, auxiliaries) from *semantic* tokens (content-bearing), and uses this distinction both in the Say-X rule and during supernode naming.

Priority	Role	Rule
1	SEM-DICT	$\text{peak_consistency_main} \geq 0.80$ AND $\text{n_distinct_peaks} \leq 1$
2	SAY-X	$\text{func_vs_sem_pct} \geq 50$ AND $\text{conf_F} \geq 0.90$ AND $\text{layer} \geq 7$
3	REL	$\text{sparsity_median} < 0.45$
4	SEM-CONC	$\text{layer} \leq 3$ OR $\text{conf_S} \geq 0.50$ OR $\text{func_vs_sem_pct} < 50$
5	REVIEW	anything not matching the above (excluded)

Table 3. Classification thresholds used for the four functional roles (§4.5). Strict priority: the first matching rule wins.

English functional vocabulary. We use a fixed list of 87 English functional tokens. The groups and their members are:

- **Copulas and copular contractions:** is, was, are, were, be, been, being, am, 's, 're, 'm.
- **Articles and demonstratives:** the, a, an, this, that, these, those.
- **Prepositions:** of, in, on, at, to, for, with, by, from, as, into, onto, upon, about, above, below, between, among, through, during, before, after.
- **Conjunctions:** and, or, but, nor, so, yet, for.
- **Relative pronouns and wh-words:** that, which, who, whom, whose, where, when, why, how.
- **Auxiliaries:** do, does, did, have, has, had, will, would, shall, should, can, could, may, might, must.

Directionality of the ± 7 -token search. When a feature peaks on a functional token, we search the probe for the nearest *semantic* target in a window of ± 7 tokens. The direction of the search depends on the functional token’s syntactic role: copulas, auxiliaries, articles, and most prepositions look *forward* (the semantic target is typically the noun phrase they introduce); *of* and possessive *'s* look *backward* (the semantic target is the head noun that governs the functional token); conjunctions and punctuation are searched *bidirectionally*, nearest-first. Multi-token semantic targets resolve to the first subword.

Cross-lingual behavior and a documented failure mode. We observed one clean cross-lingual failure during development, on a French version of the antonym seed “le contraire de ‘petit’ est”. Running the English pipeline on a French prompt produced incoherent cross-prompt activations: functional tokens were often peaked on the English contraction *'s* or on French tokens (*de*, *est*) that were not in our vocabulary, so the Say-X rule either never fired or fired on the wrong side. Adding a small French functional vocabulary and expanding the blacklist (bridging tokens that appear in both languages) restored reliable grouping. We flag this as a concrete instance of the cross-architecture-and-language limitation discussed in §7.

D. Per-Domain Full-Scale Labeled vs Random

Table 4 gives the full per-domain comparison at the standard defaults (all-three-fields, $M_{\text{ablate}} = -2$, $M_{\text{amplify}} = 20$, attention not frozen). Each domain is reported with both the labeled intervention and the matched-random-control replicate set (always exactly three replicates per labeled pair, so the random N is $3 \times$ the labeled N).

The rightmost four columns tell the same qualitative story from different angles. In the four strong domains, the labeled intervention promotes the target to a median rank (MedRk) inside the top 70 while the matched-random controls leave it near rank 200–550; in Sounds the two are indistinguishable (21 vs 24). RkGrp is the target’s rank *within the domain’s answer set* (i.e. how well the target beats its fellow answers), and is close to 1 (best) for labeled in every strong domain. Flip% reports how often the target’s logit overtakes the source at any trajectory position; it is above 96% in every strong domain’s labeled condition, which is why the operational test relies on vsMax rather than on the flip indicator alone.

Labeled–random vsMax gap by domain. Books +6.13 (Strong); USA +5.17 (Strong); Products +3.23 (Moderate); Paintings +1.58 (Weak); Sounds +0.14 (Negligible, wrong sign on vsMax comparison). The monotone relationship between this gap and the operational-usefulness verdict is one of the main reasons we report vsMax as a primary metric.

Domain	Cond.	N	Hit%	Supp%	vsMax	RkGrp	MedRk	Flip%
USA	labeled	2,450	24.7	92.8	+2.86	1.72	5	98.2
	random	7,350	0.1	83.4	−2.31	9.00	566	69.2
Books	labeled	240	3.8	69.2	+5.98	1.03	17	96.7
	random	720	0.3	75.0	−0.15	2.43	283	79.6
Products	labeled	132	15.2	65.2	+3.46	1.20	26	97.0
	random	396	0.3	87.1	+0.23	2.25	354	75.3
Paintings	labeled	90	4.4	34.4	+1.55	1.31	70	97.8
	random	270	0.0	74.1	−0.03	1.96	196	88.9
Sounds	labeled	30	0.0	100	+3.28	1.00	21	60.0
	random	90	12.2	80.0	+3.14	1.08	24	53.3

Table 4. Full-scale labeled vs. matched-random comparison at default M , all three semantic fields. N counts swap pairs (or pair-replicates for random). The labeled–random vsMax gap is the central signal of operational usefulness (Definition 2.6). Sounds inverts: matched random exceeds labeled on both Hit and vsMax.

E. Field-Additivity Detail

Each domain has three semantic fields—loosely, an *input* field (the entity being named in the prompt), an *intermediate* field (the bridging concept, e.g. the US state), and an *answer* field (the concept whose logit we measure). For each entity pair the field-additivity sweep runs seven intervention variants: three singletons, three pairs, and the all-three baseline. The field-by-field tables below give Hit%, vsMax, RkGrp, and MedRk for every variant; the bold row is the per-domain best variant chosen by Hit%.

USA (state, capital, city; 2,450 pairs/variant).

Subset	Hit%	vsMax	RkGrp	MedRk
city	2.9	−1.26	6.92	190
state	17.0	+2.38	2.72	14
capital	14.2	+0.63	3.69	11
state+capital	38.8	+4.00	1.47	3
state+city	11.1	+1.71	2.31	28
capital+city	10.2	+0.58	4.16	30
all 3	24.7	+2.86	1.72	5

Prompts are of the form “The capital of the state containing ⟨city⟩ is”. The input field is the city (what the model reads), the intermediate is the state (the reasoning bridge), and the answer is the capital. Intervening only on state and capital features produces the largest redirection: the answer’s median rank drops from 190 (city-only) to 3. Adding city-field features degrades every metric, which is the mechanism we label *less is more*: the city features encode what the model reads, not what it should produce, and their activation competes with the answer circuit.

Books (character, book, author; 240 pairs/variant).

Subset	Hit%	vsMax	RkGrp	MedRk
character	4.2	+1.87	1.77	125
book	37.1	+6.69	1.20	2
author	14.6	+4.71	1.33	12
book+author	37.1	+7.76	1.02	2
character+book	5.0	+5.22	1.23	24
character+author	3.8	+3.45	1.30	76
all 3	3.8	+5.97	1.03	17

Prompts are of the form “The author of the book featuring ⟨character⟩ is”. The character field is the input; intervening on it alone moves almost nothing. The largest single-field effect comes from the book field (37.1% Hit on its own), and adding the author field pushes vsMax to +7.76 without adding new hits. Adding the character field undoes the effect almost entirely (3.8% Hit for all-three). This is the sharpest single-domain illustration of the less-is-more effect.

Products (product, company, founder; 132 pairs/variant).

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Subset	Hit%	vsMax	RkGrp	MedRk
product	0.0	+1.90	1.61	137
company	5.3	+2.62	1.39	128
founder	8.3	+2.08	1.27	18
company+founder	24.2	+3.06	1.27	18
product+company	1.5	+2.78	1.31	93
product+founder	2.3	+2.54	1.23	48
all 3	15.2	+3.47	1.20	26

Paintings (painting, painter, first_name; 90 pairs/variant).

Subset	Hit%	vsMax	RkGrp	MedRk
painting	3.3	+1.50	1.44	71
painter	1.1	+1.68	1.30	66
first_name	6.7	+1.46	1.41	90
painter+first_name	1.1	+1.69	1.30	65
all 3	3.3	+1.55	1.32	66

Paintings is the weakest strong domain. The answer field is the painter’s first name, a coarse token with many non-painting associations (Claude, Pablo, Leonardo); and the painter supernode overlaps with `first_name` by construction (the painter supernode name already contains the first name). The less-is-more effect is correspondingly weak: the +3.4 pp main-text gap is the smallest of the four strong domains.

Sounds (sound, animal, color; 30 pairs/variant).

Subset	Hit%	vsMax	RkGrp	MedRk
sound	20.0	+5.49	1.00	5
animal	12.5	+3.49	1.21	6
color	0.0	+3.09	1.07	48
sound+animal	20.0	+4.69	1.07	5
all 3	0.0	+3.40	1.00	20

Sounds’ best variant is *input+intermediate* (sound+animal), unlike the four strong domains where the best variant is intermediate+answer. The answer field (color) contributes almost nothing on its own, and including it collapses Hit% to zero. We interpret this in §M in terms of the color answer set’s near-uniform competition.

E.1. Cross-domain single-field aggregates

Averaging over all five domains, the three field types produce very different redirection strengths. Table 5 reports the aggregate numbers: the *intermediate* field carries the strongest single-field signal (14.6% Hit), followed by the *answer* field (8.8%), with the *input* field trailing at 6.1%. Input-field features encode what the model reads; intermediate- and answer-field features encode what the model should produce.

Field type	Hit%	Supp%	MedRk	vsMax	RkGrp
Input	6.1	63.2	106	+1.90	2.55
Intermediate	14.6	79.1	43	+3.37	1.56
Answer	8.8	89.5	36	+2.39	1.75

Table 5. Per-field aggregates across all five domains. Each row averages the corresponding single-field variant across the five domains, weighted equally.

F. Adaptive M -Search

F.1. The two-phase protocol

When a swap pair misses at the default $M_{\text{amplify}}=20$, we search for a better amplifier in two phases.

Phase 1 (coarse geometric probe). We evaluate $M \in \{0.1, 0.25, 0.6, 1.5, 4.0, 10.0, 20.0\}$ in that order and stop at the first hit. The grid is geometric so that the same number of probes covers both the fine-grained low- M region where some pairs find a hit near $M \sim 2$ and the $M \approx 20$ neighborhood that the default already explored.

Phase 2 (KL-transition binary refinement). If Phase 1 produces no hit, we compute the KL divergence between the steered and unsteered output distributions at each Phase-1 grid point. The KL curve is close to linear in M for most pairs ($R^2 > 0.93$ for 8 of 10 probed USA pairs), but the hits that do exist tend to cluster near the *onset* of meaningful response rather than at arbitrary large M . We therefore identify the interval in which KL rises most steeply (the “KL transition”), place a binary-refinement search inside that interval, and run 6 additional probes, accepting a hit at any refinement point. The rationale is that the model’s response to multi-feature amplification typically has a sharp onset at a pair-specific M , and hits live near that onset.

F.2. Per-domain rescue totals

Table 6 summarizes the rescue rates split by condition (Labeled vs Field-Additivity) and domain.

Cond.	Domain	Eligible	New hits	Hit%	Before	After	Δ
Lab.	USA	870	329	37.8	24.7	38.2	+13.5
Lab.	Books	85	12	14.1	3.3	9.0	+5.7
Lab.	Paintings	103	7	6.8	4.0	9.7	+5.6
Lab.	Sounds	23	5	21.7	0.0	16.7	+16.7
Lab.	Products	128	4	3.1	13.2	15.5	+2.3
FA	USA	553	78	14.1	64.2	67.3	+3.2
FA	Books	73	9	12.3	59.5	63.8	+4.3
FA	Paintings	96	6	6.3	16.9	21.8	+4.8
FA	Products	124	2	1.6	26.4	27.6	+1.1
FA	Sounds	21	0	0.0	20.0	20.0	+0.0
Total	—	2,076	452	—	—	—	—

Table 6. Adaptive M -search rescue rates. “Eligible” is the number of pairs that missed at the default. “Before” and “After” are the domain-level Hit% before and after the search. Labeled sweeps recover between 2.3 and 16.7 pp per domain; field-additivity sweeps recover more modestly because most pairs that were going to be rescued by M -change already were, and the remaining misses tend to be feature-specificity failures rather than overshoot failures.

The overall pattern: 93% of adaptive hits are found in Phase 1 (coarse geometric), and the winning- M distribution is bimodal at $M \sim 2.4$ (47% of labeled hits) and $M \sim 6.9$ (40%), with a long tail to $M \geq 10$ (13%).

F.3. High- M rescue is null

To rule out the alternative explanation that the $M=20$ default is *too small*, we ran $M \in \{50, 100, 200\}$ on 80 near-miss pairs across three domains. The result was 0 new hits. Existing hits at $M=20$ degraded monotonically as M increased: at $M=200$ the generated continuation is typically dominated by a single high-frequency token from the amplification set, with no remaining structural coherence. KL between steered and unsteered saturates above $M \sim 50$, so the model’s output distribution is essentially maximally disrupted long before M reaches 200.

F.4. Top- k rescue is null

A second alternative is that the $M=20$ signal is distributed over too many features, and that concentrating amplification on the top- k features by graph influence might yield targeted hits at high M . On `products`→`facebook` we ran the full factorial $k \in \{1, 3, 5, 10, 67\}$ with $M \in \{20, 50, 100, 200\}$ (six source entities, so 120 runs per cell). The result was 0 hits for any $k < 67$ at any M . Only the full 67-feature set achieves hits. A worked example: at $k=1$, $M=20$ the top-influence feature for `alibaba`→`facebook` generates “the founder of Alibaba in 1999...”, at $k=3$ the output is garbled token sequences, at $k=10$ the first token is a programming-code token, and only at $k=67$ does the model correctly emit “Mark Zuckerberg...”. Graph influence correlates *negatively* with stored activation in this domain ($\rho = -0.362$, $p=0.003$): the top- k by graph influence are the loudest but not the most informative features, and concentrating amplification on them produces generic “question answering” output rather than the target answer.

F.5. Five failure modes surfaced by the M -sweep

The pairs that remain unrescued after adaptive search fall into five qualitatively different failure modes. Identifying the five modes was itself a research output: the labels below are pointers to the investigation rather than anecdotes.

(1) *Severe overshoot*. A pair whose hit exists at small M but is destroyed at the default $M=20$. Rescuable by M reduction. Example: `oklahoma_tulsa` high-vsMax pairs, 37/42 pairs rescued at $M=5$ (88% rescue rate). At $M=20$ the first token is typically a garbage token (`AddTagHelper`) followed by “Oklahoma, is Tulsa” (the city, not the capital); at $M=5$ the first token is a comma and the continuation corrects to “Oklahoma City, and Tulsa is Oklahoma”.

(2) *Field interference*. A pair that hits on the intermediate+answer subset but not on the all-three-fields variant. Rescuable by field-additivity search (the field-composition effect of §6.2.1).

(3) *Feature-interaction noise*. The label is correct but at high M distributed noise masks the target. Example: `indiana→arkansas` produces `tonode` at $M=20$ and correctly produces `Little Rock` at $M=5$. A subset of these are rescued by M reduction; the remainder are irreducible.

(4) *Signal collapse*. The target logit collapses below a pair-specific M threshold. Example: `vermont_burlington→kansas_wichita` has vsMax 8.34 at $M=20$ but only 0.31 at $M=5$. For these pairs the useful M range is narrow and lies above the default; no value of M in $\{5, 7, 10, 20\}$ produces a hit.

(5) *Feature specificity failure*. The intervention correctly activates the answer concept but the wrong specific entity within the concept. Example: `vermont→kansas` generates “Kansas is Hutchinson” (wrong Kansas city), not “Topeka”. Not fixable by M adjustment: the features that were grouped under the Kansas supernode appear to encode “Kansas cities” rather than “Topeka” specifically.

The coarse taxonomy is important for interpreting the aggregate recovery numbers. Modes (1)–(3) are rescued by adaptive M -search; mode (4) is occasionally rescued; mode (5) is not.

G. Position-0 vs Best-Over-Trajectory Metrics

We report both position-0 and best-over-trajectory variants of vsMax and Hit because they diverge in informative ways, and neither strictly dominates the other.

Domain	Lab. pos 0	Lab. best	Rand. pos 0	Rand. best
USA	52.9	91.0	2.8	13.1
Books	92.5	98.8	4.6	45.8
Products	70.5	75.0	15.2	51.5
Paintings	61.1	81.1	14.4	47.8
Sounds	50.0	90.0	63.3	93.3

Table 7. vsMax > 0 rate (%) at position 0 and at the best trajectory position. Labeled condition in the first two columns, matched-random in the last two.

A position-0 vsMax > 0 rate is a *necessary* condition for a full output success in Books and Products: in those domains no pair with a non-positive position-0 vsMax ever produces a Hit. USA is more forgiving—28.7% of USA hits have position-0 vsMax ≤ 0 and recover during generation—but position-0 vsMax still tracks output quality closely. Sounds is the one domain in which position-0 vsMax is uninformative: the matched-random controls actually exceed labeled at position 0 (63.3% vs 50.0%), and the best-over-trajectory numbers are also essentially tied. The Sounds anomaly is one of the reasons we exclude Sounds from cross-domain inference for the main operational-usefulness claim.

Books illustration. Mean position-0 vsMax is +7.29 in the labeled condition and −5.80 in matched random, a gap of +13.09 logits. The best-over-trajectory gap is smaller (+8.01), so the position-0 view nearly doubles the effect size in this domain. The reason is that Books has only 16 entities with sharply distinct author signatures: any sufficiently large intervention tends to produce *some* late-trajectory recovery of the target logit (trajectory best-over recovers for matched-random at 45.8%), but the position-0 picture cleanly separates on-topic redirection from generic recovery.

USA illustration of the less-is-more effect. In USA, the full labeled intervention has a position-0 vsMax > 0 rate of only 27.2% (median position-0 vsMax −3.88); the best field-additivity variant has 52.9% at position 0 with median vsMax

+0.31. The input-field features in the full triple cause an initial disruption that the trajectory partially compensates for, but the position-0 view makes the disruption visible.

H. KL Divergence as a Steering Diagnostic

Linearity of KL in M . Within the $M \in [5, 20]$ range, the per-pair KL curve between the steered and unsteered distributions is close to linear: $R^2 > 0.93$ for 8 of the 10 USA pairs on which we estimated the linear fit. Table 8 lists the slope and intercept of the linear fit for each probed pair; the slope varies by a factor of $5\times$ across pairs, and is strongly predicted by the pair’s ablate count (more source features ablated $\Rightarrow$ steeper KL rise per unit M).

Pair (source $\rightarrow$ target)	Slope a	Intercept b	R^2	amp.	total
kansas $\rightarrow$ oklahoma	0.318	8.28	0.941	73	163
delaware $\rightarrow$ oklahoma	0.269	7.49	0.995	73	152
texas $\rightarrow$ oklahoma	0.218	9.34	0.986	73	138
florida $\rightarrow$ oklahoma	0.263	10.06	1.000	73	139
vermont $\rightarrow$ kansas	0.079	8.43	0.864	90	130
rhode_island $\rightarrow$ wisconsin	0.194	8.16	0.982	86	156
iowa $\rightarrow$ utah	0.245	6.99	0.935	93	253
indiana $\rightarrow$ arkansas	0.374	5.09	0.967	69	269
indiana $\rightarrow$ minnesota	0.317	6.16	0.946	82	282
hawaii $\rightarrow$ oklahoma	0.109	13.69	0.848	73	158

Table 8. Per-pair linear fits of $\text{KL}(\text{baseline} \parallel \text{steered})$ as a function of M , estimated from $M \in \{5, 10, 20\}$. `amp.` and `total` count the amplify and total features.

The slope-count correlations are $r(\text{slope}, \text{ablate_count}) = +0.68$ and $r(\text{intercept}, \text{ablate_count}) = -0.63$: high-ablate-count pairs have steeper KL rises and lower starting KL, so they are the ones most sensitive to M and most likely to benefit from M -reduction.

KL ≥ 12 as a hit veto. Across 30 USA observations, no hit occurs at $\text{KL} \geq 12$. As a binary classifier on “hit iff $\text{KL} < 12$ ”, recall is $7/7 = 100\%$ and precision is $7/19 = 37\%$; the 12 false positives break down into three feature-specificity failures, three evaluator gaps (correct output miscounted due to tokenization differences such as “St. Paul” vs “Saint Paul”), three signal-collapse pairs, and three near-threshold misses. KL therefore functions as a reliable veto rather than a predictor: it reliably rules out failure modes where the output distribution is too disrupted for any target to survive, but it does not by itself predict which of the under-the-threshold pairs will become hits.

Use in adaptive search. The adaptive harness uses this in four steps: (i) run at $M=20$ and measure KL at position 0; (ii) if $\text{KL} \geq 12$, fit the linear $\text{KL}(M)$ from one additional probe and compute a pair-specific $M_{\text{crit}} = (12 - b)/a$; (iii) re-run at $M = \lfloor 0.8 \cdot M_{\text{crit}} \rfloor$ with a safety margin; (iv) if the intercept b already exceeds 12 (as for hawaii $\rightarrow$ oklahoma, where $b=13.69$), flag the pair as intrinsically disruptive and do not probe further.

A note on target recovery as a signal. An earlier version of the analysis promoted *target recovery rate*—whether the target’s logit exceeds its own unsteered baseline at any trajectory position—as a primary label-evidence metric, on the basis of a 92% (labeled) vs 29% (random) gap in USA regime C. A deliberate replication attempt on Books collapsed this gap to 92% vs 89% in the same regime, because the Books model’s smaller answer set and distinctive author signatures make random recovery mechanically easy. The lesson, which we took seriously when finalizing the main-paper metrics, is that high-discrimination binary flags may reflect a single domain’s structure; *continuous* magnitude metrics (max excess over baseline, vsMax) are more portable. We therefore use Target Recovery only as one of three supporting signals in the within-regime-C analysis of Section 6.1, not as a primary metric.

I. Cross-Prompt Robustness at Scale

We ran a scalable cross-prompt analysis on 1,607 intra-domain entity pairs across 5 domains (103 entities; 5,000 bootstrap resamples for Jaccard CIs and 2,000 permutations per pool size for significance).

Jaccard overlap with bootstrap CIs. Table 9 gives the mean intra-domain Jaccard overlap between entity feature sets, together with a 95% bootstrap CI and the pair count. All five domains have overlaps significantly above chance ($p < 0.001$) at every tested pool size.

Domain	Jaccard	95% CI	N pairs
USA	0.465	[0.462, 0.468]	1,225
Books	0.308	[0.302, 0.315]	210
Products	0.364	[0.356, 0.374]	91
Paintings	0.286	[0.279, 0.292]	66
Sounds	0.621	[0.597, 0.648]	15

Table 9. Intra-domain Jaccard overlap between per-entity feature sets, with bootstrap 95% CIs.

Activation stability. Activation stability measures whether a feature’s activation pattern is consistent across the generated probe set for a given entity. Stability is high in every domain: USA 0.947, Books 0.908, Products 0.903, Paintings 0.916, Sounds 0.944. Peak-token agreement is 85–98% and peak-type (functional vs semantic) agreement is 93–99%.

Layer-binned overlap and the early/late ratio. The richer signal is layer-binned. Table 10 reports Jaccard overlap separately in early, middle, and late layer bins. Early layers host a shared “backbone” whose features are reused across entities (high overlap), while late layers host entity-specific features (low overlap). The ratio of early to late overlap is $1.4\times$ – $3.0\times$ in four of five domains, consistent with a backbone-and-specialization reading.

Domain	Early	Middle	Late	Early/Late ratio
USA	0.543	0.440	0.293	$1.85\times$
Books	0.347	0.340	0.184	$1.89\times$
Products	0.496	0.308	0.164	$3.02\times$
Paintings	0.302	0.311	0.212	$1.43\times$
Sounds	0.684	0.544	0.440	$1.56\times$

Table 10. Layer-binned intra-domain Jaccard overlap and early/late ratio.

Supernode consistency across entities. A complementary view: for each feature that appears in two entity graphs, is it assigned to the *same* supernode (*scaffold*), *regrouped* into a different supernode (e.g. *Say* (Austin) in the Dallas graph vs *Say* (Sacramento) in the Oakland graph), or is the assignment *inconsistent* (same feature, semantically unrelated supernodes)? Table 11 gives the breakdown.

Domain	Same (%)	Regrouped (%)	Inconsistent (%)
USA	76.6	16.2	7.3
Books	47.5	12.8	39.7
Products	65.2	16.5	18.4
Paintings	71.0	9.9	19.1
Sounds	85.6	8.9	5.6

Table 11. Supernode consistency across entity graphs within each domain. “Same” is the scaffold; “Regrouped” captures entity-appropriate re-assignment (typically *Say-X* features pointing at the new answer); “Inconsistent” flags same-feature-different-role mismatches.

Caveats. We note four non-trivial caveats for this table. First, the original Dallas/Oakland pair that motivated the early analyses is a ~ 93 rd-percentile outlier within USA; population means are 10–15 pp lower than the single-pair numbers originally reported. Second, the Books “Inconsistent” rate (39.7%) likely reflects keyword-detection limits on literary names (the matcher’s treatment of “Anna Karenina” and similar) rather than genuine grouping instability. Third, Paintings’ weak/inverted layer gradient is unexplained and may be driven by the small- N and high-error-node-rate structural issues discussed in §M. Fourth, Sounds has the highest overlap but zero Hit rate at default M , which is the correct message that high Jaccard overlap is necessary but not sufficient for successful swap.

J. Graph Scaffold Analysis

J.1. Definitions

We partition each cross-entity feature population into three groups. *Scaffold features* appear in both graphs with the *same* supernode assignment (usually structural primitives like copulas, prepositions, and task operators). *Regrouped features* appear in both graphs but with different supernode names (typically Say-X features that point at the new answer, e.g. Say (Austin) vs Say (Sacramento)). *Entity-only features* appear in one graph only. *Scaffold influence* is the fraction of the graph’s total influence carried by scaffold features.

J.2. A worked example: Dallas vs Oakland

The reference pair that originally motivated the scaffold metric is Dallas (Texas capital problem) vs Oakland (California capital problem). Table 12 decomposes it.

Population	N features	Dallas infl.	Oakland infl.
Scaffold (shared + same supernode)	119	50.6%	49.2%
Regrouped (shared + different supernode)	29	12.3%	13.7%
Entity-only	55/62	21.7%	19.5%

Table 12. Dallas/Oakland scaffold decomposition. Scaffold features account for roughly half the total graph influence in both graphs, and their influence is concentrated in early layers (0–5, where they dominate 75–100% of total influence).

Of the 29 regrouped features, 27 are entity-appropriate re-assignments: 15 Say (Austin) $\rightarrow$ Say (Sacramento), 4 Texas $\rightarrow$ California, 3 Dallas $\rightarrow$ Oakland, 2 Say (Texas) $\rightarrow$ Say (California), and 1 Austin $\rightarrow$ Sacramento. Only 4 of the 29 (roughly 2.7% of the shared-feature population) are genuinely inconsistent regroupings. This is the pattern we formalize at the population level in Table 11.

J.3. Cross-domain scaffold gradient

At the domain level, scaffold influence perfectly orders the four strong domains by Hit% at the all-three-fields labeled baseline (Spearman $\rho=1.0$, $N=4$).

Domain	Scaffold infl.	Shared%	Early scaffold	Late scaffold	Hit%
USA	0.530	63.7%	74.6%	16.3%	24.7
Products	0.422	52.1%	62.1%	10.3%	15.2
Paintings	0.359	44.8%	50.9%	9.2%	4.4
Books	0.253	48.9%	39.7%	4.3%	3.8

Table 13. Per-domain scaffold metrics. “Scaffold influence” is the fraction of total graph influence carried by scaffold features; “Shared%” is the fraction of features (by count, not influence) that are scaffold; “Early” and “Late” are the same quantity restricted to layers $\leq L_5$ and $\geq L_{16}$ respectively.

The late-layer gradient is the most striking element of this table: USA shares 16.3% of its late-layer influence as scaffold, while Books shares only 4.3%. The output-generation layers are where entity-specific features concentrate, and where structural compatibility matters most. This connects mechanistically to the less-is-more effect: interventions that include input-field features disrupt the fragile late-layer scaffold.

J.4. Within-domain pair-level prediction is null in USA

The within-domain version of the scaffold test is considerably weaker. Table 14 splits the 2,450 USA pairs into four quartiles of pair-level scaffold compatibility and reports the Hit% and mean vsMax for each quartile.

The within-USA scaffold range is $[0.38, 0.73]$, concentrated around 0.53, and the quartile boundaries fall within a 0.12-wide window. This is almost certainly too narrow to discriminate pairs. In smaller domains, the within-domain picture is noisy: Paintings has a target-residualized correlation of $r = +0.334$, Products $r = +0.29$ ($N=132$), and Books $r = -0.06$ ($N=240$, fragile at 9 hits). We therefore treat the scaffold claim as a strictly between-domain result.

Scaffold quartile	Mean scaffold	Hit%	Mean vsMax
Q1 (lowest)	0.468	27.8	+3.15
Q2	0.513	23.2	+2.90
Q3	0.545	23.7	+2.94
Q4 (highest)	0.593	24.3	+2.43

Table 14. USA quartile analysis of pair-level scaffold compatibility. No monotone trend; the highest-scaffold quartile does not have the highest Hit%.

K. Logit-Shift Regime Taxonomy

K.1. Definitions (position 0)

We classify every swap at position 0 by what happens to the target and source logits relative to their unsteered baselines. Regime **A** is “target UP, source DOWN, flip” (clean redirection). Regime **C** is “both DOWN, flip” (differential disruption in which the target is less disrupted than the source). Regime **D** is “both DOWN, no flip” (generic disruption). Regime **E** is “target FLAT, source DOWN” (pure suppression with no target promotion).

K.2. Prevalence at the default and at the best variant

Table 15 compares regime prevalence across three conditions: the per-pair best field-additivity variant, the full labeled ($M=20$, all three fields), and matched-random. The clean-redirection regime A is the one we ultimately want to maximize; the generic-disruption regime D is the one matched-random concentrates in.

Domain	Regime A (%)			Regime D (%)		
	best FA	full lab	random	best FA	full lab	random
USA	34.9	8.9	19.4	9.1	19.4	45.3
Books	62.1	38.8	40.8	3.3	3.3	34.6
Products	62.1	56.8	51.5	2.3	2.3	22.0
Paintings	47.8	17.8	23.3	2.2	6.7	42.2

Table 15. Regime A (clean redirection) and Regime D (generic disruption) prevalence under the per-pair best field-additivity variant, full labeled, and matched-random. The best variant dramatically amplifies regime A and suppresses regime D compared to full labeled.

Three patterns stand out. First, the best FA variant pushes many more cases into regime A than the full labeled variant does (USA: 8.9 $\rightarrow$ 34.9%; Books: 38.8 $\rightarrow$ 62.1%); removing input-field features eliminates generic disruption that was pushing cases into regime C/D. Second, hit rates *within* regime A also improve: USA regime A hits go from 31% (full labeled) to 46% (best variant); Books regime A hits go from 4.3% to 54.4%. The best variant is not just redistributing cases across regimes, it is improving the quality of each regime. Third, regime D nearly vanishes in the labeled condition (19.4 $\rightarrow$ 9.1% in USA; 6.7 $\rightarrow$ 2.2% in Paintings), while matched-random concentrates in regime D (42–45% in USA and Paintings).

K.3. Within-regime-C signals (USA)

Regime C is the largest and most ambiguous regime in the labeled condition: both logits drop, but the target less so. The richest within-regime-C signals are given in Table 16, which is the data behind the within-regime-C result reported in Section 6.1.

Signal	Labeled	Random
Target-recovery rate (%)	92.2	29.3
Sustained dominance (tgt_win_pct)	0.673	0.319
Mean vsMax	+2.33	−0.10
Hit% (conditional on regime C)	22.5	0.6

Table 16. Within-regime-C signals, USA. Labeled features cause differential disruption in which the target recovers above its own unsteered baseline while the source stays suppressed; matched-random does not.

The Target-Recovery signal is strong in USA regime C but is domain-dependent: the same measurement in Books regime C

gives 91.9% (labeled) vs 88.9% (random), essentially no gap. The reason is that Books has a small answer space and sharply distinct entity signatures, so mechanically *any* sufficiently large perturbation tends to recover the target above its baseline. We therefore report Target-Recovery only as one of three supporting within-regime signals rather than as a primary label-evidence metric.

L. Case Studies

This section collects worked case studies that would be too discursive for the main text. Each entry gives the source and target entity, the field-additivity variant used, the intervention size, the vsMax achieved, and the first ~ 10 generated tokens of the model’s continuation.

L.1. Six success cases

USA: Colorado Springs $\rightarrow$ Detroit. Source prompt: “The capital of the state containing Colorado Springs is”; target entity: Detroit (Michigan, capital Lansing). Best variant: `state+capital`. vsMax +17.5, target logit rank moves from 342 to 1. Generated continuation: “... Lansing, the capital of Michigan...”. Textbook regime A: target UP, source DOWN, clean flip, on-topic continuation.

Books: Holden Caulfield $\rightarrow$ Katniss Everdeen. Source prompt: “The author of the book featuring Holden Caulfield is”; target entity: Katniss Everdeen (The Hunger Games, author Suzanne Collins). Best variant: `book+author`, 11 ablate features and 63 amplify features. vsMax +21.79, the largest single-pair vsMax in our dataset. Generated continuation: “... Suzanne Collins, who wrote The Hunger Games...”. This pair illustrates how small a correct amplify set can be and still dominate a $50\times$ -larger graph.

Products: Nike $\rightarrow$ Tesla. Source prompt: “The founder of the company that makes Nike sneakers is”; target: Tesla (Elon Musk). Best variant: `company+founder`. vsMax +12.25, target rank 5,933 $\rightarrow$ 1. Generated continuation: “Elon Musk is the founder...”, with the steered first-token probability $p=0.82$. The baseline (un-steered) output is “Phil Knight...”; the redirection is complete at position 0.

Paintings: La Grande Jatte $\rightarrow$ Water Lilies. Source: “The painter of La Grande Jatte is”; target: Water Lilies (Monet). Best variant: `painter+first_name`. vsMax +9.375. This is the only Hit in the full Paintings batch (1/90 pairs). The target’s baseline rank for “Claude” was already 3, and the intervention converts this into a first-position hit. We include this pair to illustrate that Paintings can be redirected, but only when baseline conditions are already favorable.

USA with all three fields: Minnesota $\rightarrow$ Florida. Source: “... Minnesota”; target: Florida (Tallahassee). With all three fields and $M=20$: vsMax +16.06 but `flip@0` is `False`, and the control-stability measure reaches 16.89 (very high collateral disruption). The target logit does recover along the trajectory, but the initial disruption is severe and the generation starts with an incoherent token. This is a clear illustration of the all-fields cost: the intervention has a large vsMax yet no Hit, because the position-0 disruption dominates the first generated token.

Sounds: Bark $\rightarrow$ Hoot (mechanical artifact). Source: “The color of an animal that barks is”; target: an owl that hoots. Both answers resolve to “brown”. The intervention “succeeds” in the Hit sense (the target token “brown” appears in the continuation) but only because the source and target *share* the answer. This is not evidence of label quality—it is the structural confound that causes Sounds to fail the matched-control condition (Definition 2.6).

L.2. Seven failure cases covering the five failure modes

Kansas/Wichita $\rightarrow$ New Hampshire/Manchester (total failure, feature specificity). vsMax -6.875 , target rank 5,816, first token “Oklahoma” and continuation “... Oklahoma City, which is the capital...”. The Kansas source features are successfully suppressed, but the amplified New Hampshire features do not encode Manchester-capital circuitry strongly enough to dominate the output. Failure mode: feature specificity.

Utah/Provo $\rightarrow$ Iowa/Cedar Rapids (rank-hit misalignment). vsMax +14.0, RkGrp 1, but the generated continuation is garbled (not a clean Des Moines production). The target logit wins at position 0 but the subsequent tokens are disrupted enough that the Hit criterion (target token anywhere in continuation) fails. Failure mode: feature-interaction noise.

Ohio/Cleveland → Oklahoma/Tulsa (regime D). vsMax +13.31 but flip@0 is False. The target logit rises sharply along the trajectory but never overtakes the source at position 0, and the autoregressive decoder never recovers. At $M=5$ this same pair becomes a Hit. Failure mode: severe overshoot, rescuable by M -reduction.

Scout Finch → Huckleberry Finn (field interference / substring confound). 176 amplify features (the highest count in Books), and a substring confound between the two targets' names (Finn is a common suffix). Generated continuation is garbled. The pair is rescuable at the book+author variant but not at the all-three-fields baseline. Failure mode: field interference.

Windows → Oculus (product-identity collapse). The source (Windows/Microsoft) is successfully suppressed but the generated output is "Rift" (the Oculus product) rather than "Palmer Luckey" (the founder, our target). The "company" supernodes in the Products domain encode product identity, not founder identity, and amplifying them therefore promotes the product name rather than its founder. Failure mode: feature specificity—a case where even a perfectly-executed intervention targets the wrong entity because the label's extension does not match the target concept.

Girl with a Pearl Earring → Guernica (cross-concept collapse). 195 amplify and 88 ablate features fail to dislodge Vermeer (the source's painter) in the continuation; the target "Pablo" ends up at rank 1,382 at position 0. This is a case where the source entity's circuit is deeply entrenched (Vermeer features are shared across the painting and its description) and the target entity is coarse (Picasso has many broader associations). Failure mode: feature specificity combined with low scaffold compatibility.

Hiss → Hoot (thin-margin collapse in Sounds). vsMax -0.5; the intervention nudges the target color "brown" to rank 2, while "black" takes rank 1. In a $K=6$ answer space with near-uniform logit competition this is a mechanical artifact: any sufficiently large perturbation moves the winner by one rank, and half the time the wrong color wins. Failure mode: thin margins (see §M).

L.3. Suppression-only zone

A large fraction of swaps—the single largest outcome category in every domain—are suppressed without successful retargeting: the source is absent from the output but the target is not present either. These are USA 59.2%, Books 48.3%, Products 46.2%, Paintings 68.9%, Sounds 66.7%. This is the main-paper observation that *suppression is generic; targeting requires specific features*.

L.4. Threats to validity for case studies

Case studies select for extremes (the largest vsMax pairs and the most striking failures), and single-example interpretations should not be over-generalized. The distribution-level claims live in the full labeled-vs-random tables (§D), field-additivity tables (§E), and adaptive- M rescue rates (§F). Post-hoc explanations of success and failure (e.g. "the company supernodes encode product identity") are hypotheses supported by the particular pair, not conclusions.

M. Negative-Result Domains: Sounds and Paintings

M.1. Why Sounds fails the matched-control condition

Sounds is the one domain in which labeled supernodes fail the operational-usefulness test. Three structural causes converge, and we have direct evidence for each.

Small N and shared answers. The Sounds domain has only 6 entities, so the non-identity pair set has size 30. Of these 30 pairs, 12 share an answer (*brown*): different animals that make different sounds still map to the same color. A matched-random intervention that suppresses the source color also "produces" the target color simply because that color is already the most common answer in the suppression residual. This inflates the matched-random Hit% in Sounds to 12.2%, higher than the labeled 0.0%.

Tiny answer set with near-uniform competition. We measured the geometry of the Sounds answer space directly using Gemma-2-2B-it and compared it to USA and Books. Three quantities are relevant: the input-embedding pairwise cosine

(are the answer tokens close in the model’s embedding space?), the last-hidden-state pairwise cosine at the seed prompt (are the prompts confusable?), and the restricted next-token distribution over the domain answer set (how peaked is the model’s answer?). Table 17 gives the numbers.

Domain	K	Input cos	Last-hidden cos	Restricted entropy
Sounds (colors)	6	0.312	0.624	0.725
USA (capitals)	50	0.209	0.874	0.663
Books (authors)	15	0.178	0.812	0.449

Table 17. Answer-space geometry. “Input cos” is the mean pairwise cosine of mean-pooled answer token embeddings. “Last-hidden cos” is the mean pairwise cosine of the last-layer hidden state at the seed prompt, across entity prompts. “Restricted entropy” is the normalized entropy of the next-token distribution restricted to the domain answer set and renormalized.

Colors have the highest input-embedding cosine of the three domains (0.312 vs 0.209/0.178), so the six answer tokens already point in similar directions at the first layer. More importantly, the model’s restricted distribution on Sounds prompts has very high normalized entropy (0.725), which concretely means the model’s probability mass on the answer set looks like [44, 15, 13, 12, 9, 7]%; roughly four colors are live competitors, with the gap between the intended target and the next-best color only about 0.15 in probability. A steering intervention of $M=20$, tuned for USA (50-wide distribution with wider margins), is a blunt instrument on this knife-edge competition. A fine-grained M -sweep on Sounds ($M \in \{0.1, 0.2, \dots, 5.0, 20\}$) recovered up to 16.7% Hit at $M=5$, confirming that the default M overshoots the thin margins; but the domain remains statistically under-powered.

Interestingly, sounds prompts themselves are well-separated. The last-hidden-state cosine is *lower* for Sounds (0.624) than for USA (0.874) or Books (0.812). The model can distinguish the six animals perfectly well internally—the bottleneck is at the output layer, not in the middle.

Falsification verdict for Sounds. The matched-control vsMax gap on Sounds is +0.14 in favor of *random*, and the Hit-rate gap is +16.7 pp in favor of random. Sounds fails Condition 2 of Definition 2.6. We exclude Sounds from cross-domain inference for the main claim and report it as a directional negative.

M.2. Why Paintings is a weak strong domain

Paintings sits at the bottom of the four strong domains with a full-labeled Hit% of 4.4 and a vsMax gap of +1.58 (“Weak”). Three structural factors contribute.

(a) *The answer field is coarse.* The answer is the painter’s first name—Claude, Pablo, Leonardo—tokens with many non-painting associations in the general language model distribution. Amplifying features that peak on “Claude” in a Monet circuit therefore moves the logit toward a cluster of non-painting “Claude” meanings.

(b) *The painter supernode subsumes the first-name supernode.* “Monet” as a string contains much of “Claude” as a concept (the painter is the agent that the first name names), so the painter and first_name supernodes overlap by construction. This collapses the field-additivity structure: painter+first_name is not a very different intervention from painter alone.

(c) *High error-node rates.* Paintings has the highest error-node rate of any domain (17%; an error node is a node whose attribution could not be resolved during graph generation), so a substantial portion of the circuit is simply not visible to the grouping or the intervention. The maximum number of features a paintings intervention can operate on is therefore smaller than in the other domains, and the signal-to-noise ratio is correspondingly worse.

Concretely, 89/90 pairs fail at the painter+first_name variant. The only hit (La Grande Jatte → Water Lilies) already had a baseline rank of 3 for “Claude”, so the intervention only had to move the logit by two positions. We therefore treat Paintings as a Weak-strength supporting domain rather than a confirmatory one. The four-domain rank-perfect scaffold prediction (Table 13) holds without Sounds and is *consistent* with the Paintings Weak verdict.

N. Reproducibility Manifest

N.1. Code, data, demo (anonymized for review)

All artifacts are released anonymously for the review period. Repository: <https://anonymous.4open.science/r/attribution-graph-probing-anon> (commit hash and license in README.md). Interactive demo: <https://anonymous.4open.science/r/attribution-graph-probing-demo-anon>. License: GNU GPL v3. All URLs will be de-anonymized at camera ready.

N.2. Per-domain CLI (USA example)

```
# 1. Generate attribution graph (Neuronpedia API)
python scripts/00_neuronpedia_graph_generation.py \
  --model gemma-2-2b-it \
  --prompt "The capital of the state containing Dallas is" \
  --target "Austin" \
  --node-threshold 0.8 --edge-threshold 0.85 --max-nodes 5000 \
  --output_dir output/usa/dallas_austin/

# 2. Select features (interactive UI)
streamlit run eda/threshold_selection.py --tau 0.95

# 3. Probes + activations + grouping
python scripts/01_probe_prompts.py \
  --graph output/usa/dallas_austin/graph.json \
  --output_csv output/usa/dallas_austin/activations.csv
python scripts/02_node_grouping.py \
  --input output/usa/dallas_austin/activations.csv \
  --graph output/usa/dallas_austin/graph.json \
  --output output/usa/dallas_austin/grouped.csv

# 4. Run swap protocol with matched random control + adaptive M-search
python scripts/03_swap_full.py --domain usa \
  --m_ablate -2 --m_amplify 20 --freeze-attention false \
  --random-replicates 3 --random-seed 42 \
  --field-additivity true --m-search adaptive \
  --output_dir output/usa_states_batch/
```

N.3. Runtime profile

Per single-circuit pipeline (graph + activation + grouping + subgraph): 14–24 minutes on a single L4 GPU. Full per-domain swap sweeps: USA 17h, Books 1.6h, Products 0.9h, Paintings 0.6h, Sounds 0.2h on 8 H100 GPUs (matched-control replicates parallelized).

N.4. Artifact release manifest

For each of the five domains, the release contains: (a) attribution graphs in Neuronpedia JSON format; (b) the probe-prompt files; (c) activation matrices; (d) supernode groupings (pre- and post-stability filter); (e) swap dumps with per-pair Hit, vsMax, RkGrp, Sup, Flip, CtrlS, and KL trajectories; (f) field-additivity per-variant tables; (g) adaptive M -search logs (Phase 1 and Phase 2); (h) random-control sha256 seeds.

N.5. Determinism and resume

The pipeline is fully deterministic. Matched-random controls are sha256-seeded from (run_seed, pair_id, replicate, mode); generation uses $T=0.3$, $n=10$, frequency penalty 2.0, seed 42, with no random initialization elsewhere in the harness. Per-feature checkpoints permit resumption from partial runs, which is how we recover from transient Neuronpedia API failures.

O. Glossary and Notation

CLT Cross-Layer Transcoder ([Ameisen et al., 2025](#)).

CPAS Cross-Prompt Activation Signature (Table 2).

τ Cumulative-influence threshold for feature selection (default 0.95).

$M_{\text{ablate}}, M_{\text{amplify}}$ Multipliers for source and target feature decoders (defaults $-2, 20$).

vsMax $\max_{t \leq 10} (\ell_{e_B}(t) - \max_{e' \neq e_B} \ell_{e'}(t))$, where $\ell_e(t)$ is the logit of entity e 's first-subword token at trajectory position t .

Hit The first-subword token of the target's answer appears in the generated continuation.

RkGrp Best target rank within the domain answer set, minimized over trajectory positions.

Target Recovery Binary indicator: the target's logit exceeds its own unsteered baseline at some trajectory position.

Regime A/C/D/E Position-0 logit-shift regimes (§K).

Scaffold Features that appear in two entity graphs with identical supernode assignment.

Scaffold influence Fraction of total graph influence carried by scaffold features.

Field-additivity (FA) The 7-variant ablation of single-field, two-field, and three-field intervention subsets.

KL $\text{KL}(\text{baseline} \parallel \text{steered})$ at position 0, unless otherwise stated.